

IMPACT OF SOCIAL MEDIA EXPOSURE ON ACADEMIC PERFORMANCE



Project Report submitted towards the partial fulfillment for the short internship Course as part of UG Curriculum

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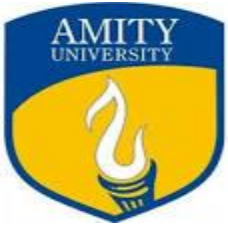
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CERTIFICATE

This is to certify that Mr. Kaushlendra (Enrolment Number: A8979121008) has completed the research for the Bachelor of Statistics degree from the Department of Statistics, Amity School of Applied Sciences, Amity University Uttar Pradesh, Lucknow, on "Impact Of Social Media Exposure On Academic Performance." Dr. Shailja Pandey, Assistant Professor, Department of Statistics, Amity University Uttar Pradesh, Lucknow, oversees its completion. Dr. Asita Kulshreshtha, Head of Institution, Department of Statistics, Amity School of Applied Sciences, Amity University Uttar Pradesh, Lucknow, and Dr. Gunjan Singh, Assistant Professor, Department of Statistics, Amity University Uttar Pradesh, Lucknow. The dissertation represents the results of the student's own original research and study, and its contents do not serve as the foundation for the awarding of any other degree to the candidate or anyone else

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ABSTRACT

Over the last few decades, social media use has grown significantly. It has become quite popular for student communication as a result of this expansion. In fact, these social media platforms can be a useful way for kids to communicate with their teachers and with one another. But excessive social media use can harm students' academic performance and cast doubt on this practice. By surveying students, this study attempts to look into the advantages and disadvantages of social media use on academic achievement. The poll investigated the most popular student social network and the one most beneficial to their academic abilities. The poll garnered 50 replies, and descriptive data demonstrate the correlation between the student's academic achievement and the amount of time spent on social media. The findings of this study can be applied to the development of a practical strategy for raising students' academic performance through better usage of online media. There were 50 responses to the poll, and the descriptive statistics show a connection between the students' academic success and their use of social media. The results of this study can be used to create a workable plan for improving students' use of internet media in order to improve their academic achievement.

Keywords: Social media networks; Academic Performance.

OBJECTIVES AND QUESTIONNAIRE

1. Which social media platform is most widely used by students?
2. How frequently do you use social media each day?
3. How might social media networks help students and teachers communicate better and build stronger relationships?

4. How might the social media platform enhance the lecture's educational content?

5. Is using social media beneficial for the educational process

RELATED WORK:

To acquire a thorough knowledge of the research, this section provides an overview of the pertinent literatures on social media use in the educational society.

CHAPTER 1

INTRODUCTION

Since the advancement of internet technology, the use of social media has experienced significant expansion. They become very well-liked and have a significant impact on all facets of our life, but particularly on schooling. For this reason, social media use and its effects on educational operations have received increased attention from researchers during the past 10 years. In [1], writers demonstrate that social media use serves a variety of functions, including facilitating the art of learning, reaching out to students, assisting students in creating their own communities in order to collaborate with one another, and sharing ideas with educational personnel.

presents the findings of the data analysis. The paper is concluded in section five, which provides additional explanation of the findings.

Additionally, our research uses both paper and the internet to collect as many student responses as possible, which is a key component of our publication's uniqueness. The purpose of this research study is to gather data on social media usage and how it affects academic achievement. The format of this essay is as follows.

We review the literature and work that is linked to our research in part two. The research technique is then described in section three, and the data analysis and findings are discussed in section four. The paper is concluded in section five, which provides additional explanation of the findings.

The evolution of social media:

Web-based applications that enable people to produce, share, or exchange knowledge, concepts, images, and videos are referred to as social media. Facebook, Twitter, YouTube, and Instagram are some of the most widely used social media platforms. The numbers provided in [7] that show Facebook had 3 million monthly reach averages and Twitter had 500 million active users globally in 2016 demonstrate the enormous expansion of social media. The entire student body is continually logged onto social networking sites. The

age range of the most frequent users of online social media is between 18 and 29 years old for teenagers and college students [8].

CHAPTER 2

DATA COLLECTION AND METHODOLOGY

A questionnaire distribution was employed in this study to adopt a quantitative methodology. In this study, we asked students their opinions on using social media and how that affected their learning. We also inquired about their daily social media usage and the social network they believed to be most useful for advancing education. The questionnaire has two parts. The first one concerns the respondents' individual data, such as their gender, educational history, and field of study. The second section of the survey includes various questions about social media usage..

Actually, the purpose of the survey is to provide information on the following questions:

Which social media site is most popular among students?

R2: How many times a day do you use social media?

R3: How may social media platforms improve the educational process of the lecture?

R4: How might social media networks enhance the lecture's instructional process?

R5 Are there any benefits to using social media in the classroom?

This study used a cross-sectional survey to assess how students' use of social media affected their academic achievement. Google Forms was used to create a questionnaire. Students' emails and WhatsApp groups were used to collect data, which was then put through a spreadsheet for analysis.

CHAPTER 3

3. RESULT AND DISCUSSION

This section presents and analyzes the data that was gathered. There are 158 respondents, 91 (57.6%) men and 67 (42.4%) women, who represent different age groups and socioeconomic classes.

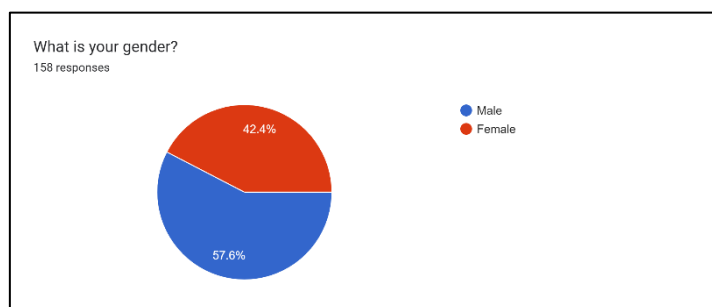


Figure 1.1 Gender

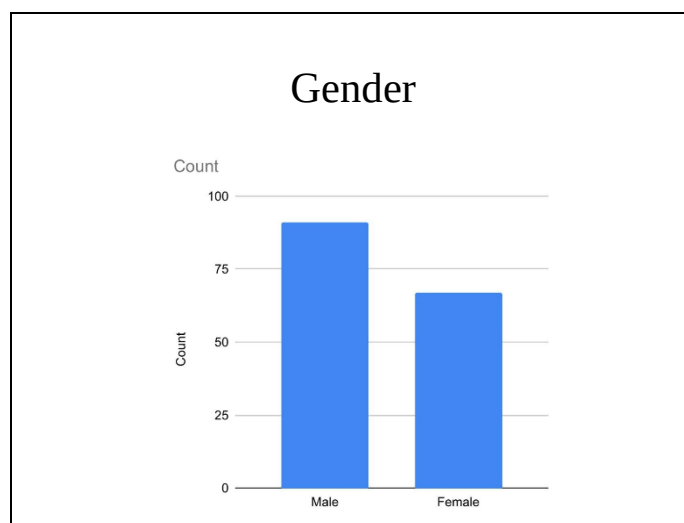


Figure 1.2 Gender

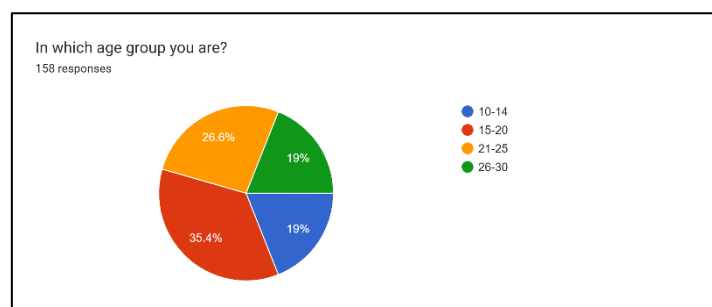


Figure 2.1 Age group

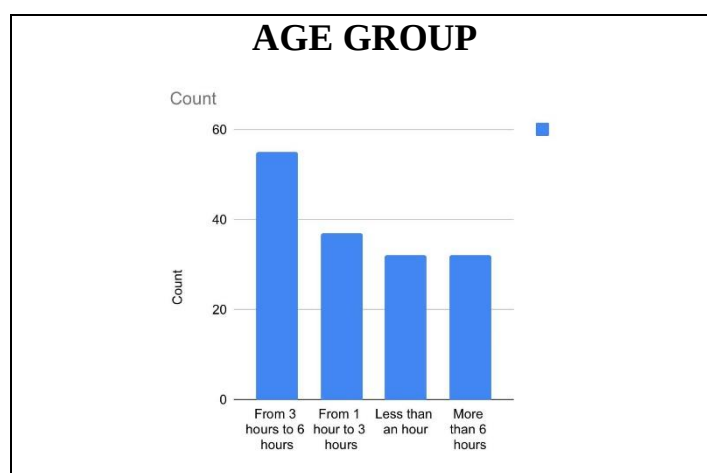


Figure 2.2 Age group

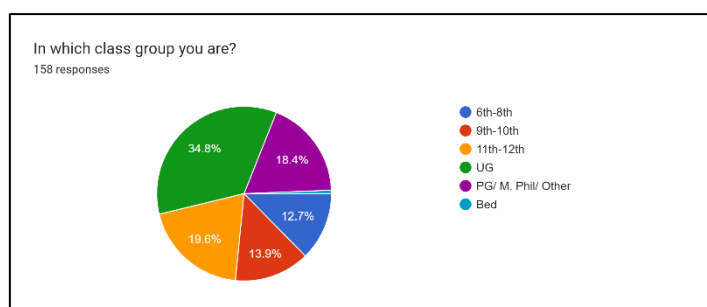


Figure 3.1 Class group

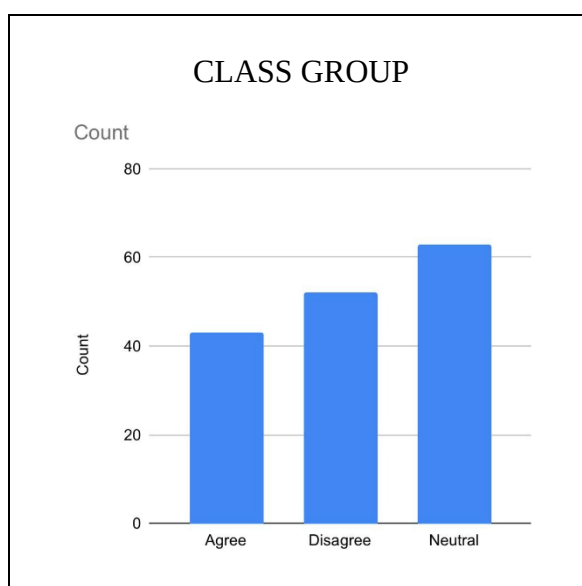


Figure 3.2 Class group

3.1. R1 – Which social media platform is most widely used by students?

Figure 4.1 shows the total number of respondents from different social networks. YouTube is the most used website for student communication, followed by Facebook and Twitter, and then other social media platforms including LinkedIn, Instagram, and What's up. 20.9 percent of the 158 respondents use Twitter, whereas 37.3% of the respondents communicate with one another on YouTube. According to the research, YouTube is almost twice as popular among students than Twitter.

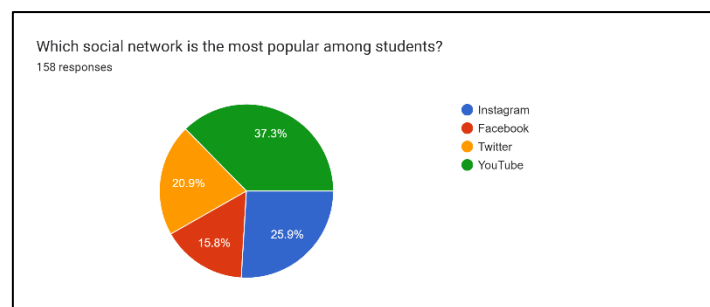


Figure 4.1 Popular social network among students

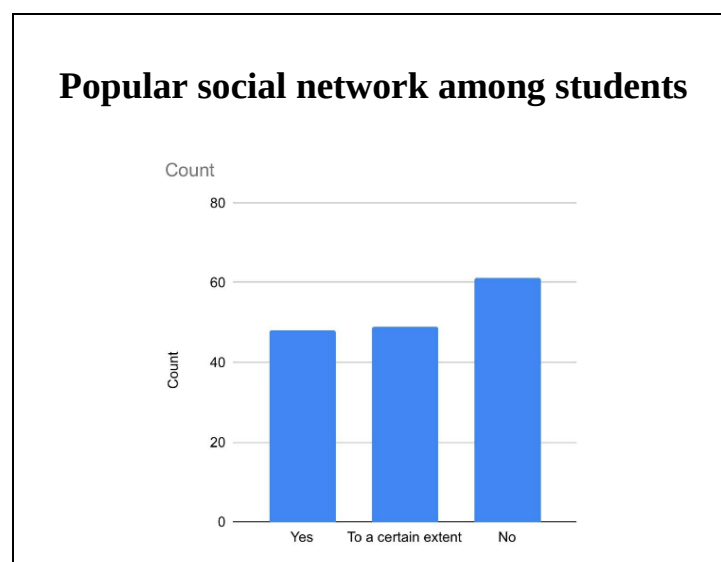


Figure 4.2 Popular social network among students

3.2. R2 – How frequently do you use social media each day?

The use of social media was evaluated using two questions. The questions are as follows:

How much time do you spend daily on social networking sites?

- Do you like to use social media only during particular times of the day?

According to the results, 23.7% of those surveyed claimed to spend one to three hours each day on social media. Figure 5.1 illustrates that 20% of respondents believe that less than an hour is sufficient for them to check our social media pages.

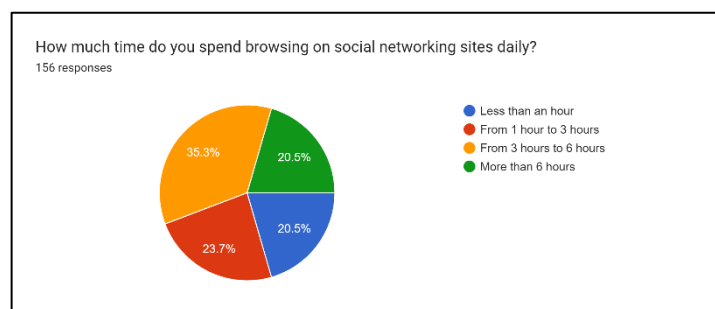


Figure 5.1 daily time spent surfing social media

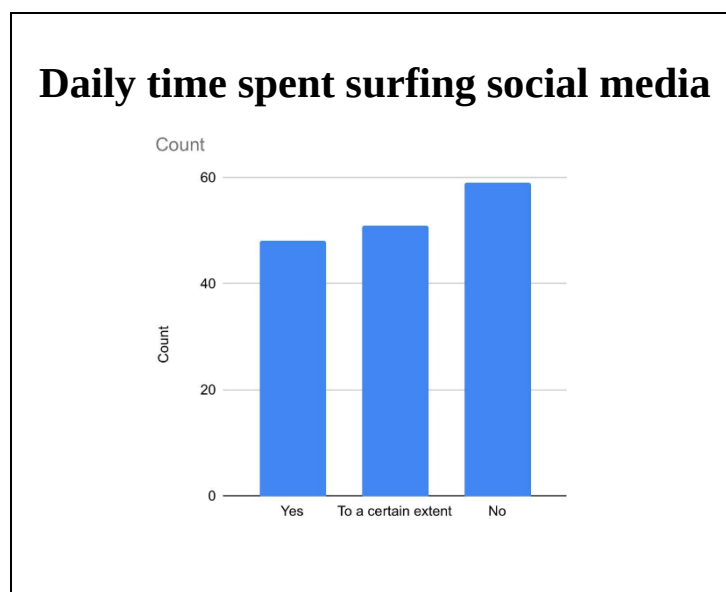


Figure 5.2 daily time spent surfing social media

Figure 6.1 shows that a sizable proportion of respondents (27.2%) agree that there should be a designated time of day for utilizing social media. 39.9% are neutral, while 32.9 percent are against the idea and would rather see it utilized continuously.

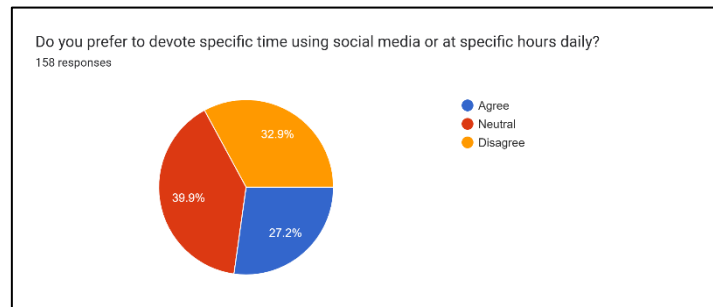


Figure 6.1 reserving a specific amount of time each day for social media use

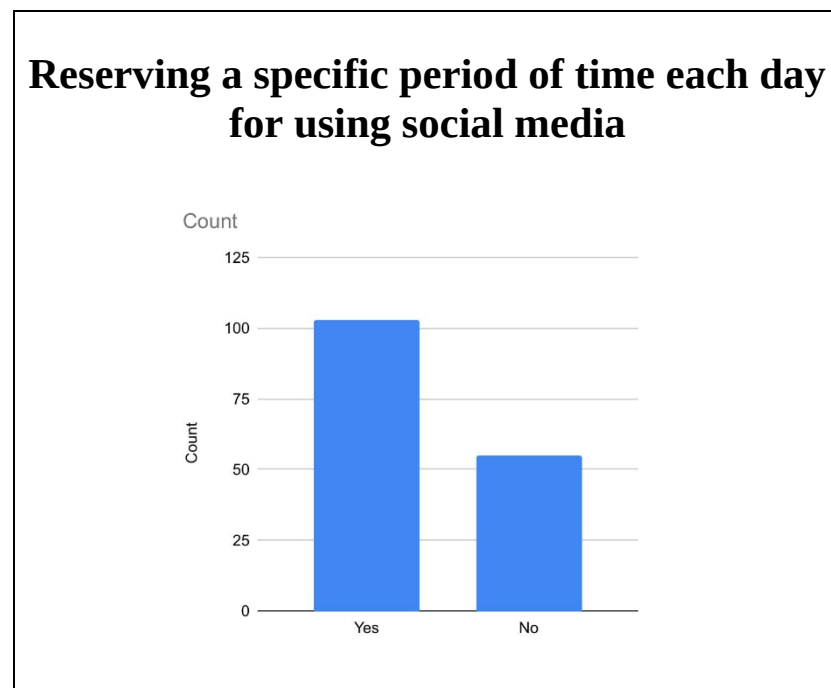


Figure 6.2 reserving a specific period of time each day for using social media

3.3. R3 – How may social media networks enhance the lecture's instructional process?

The three questions listed below were used to assess how social media has enhanced communication between students and teachers:

- Do the instructors grant you access to their online profiles on social media?
- Do teachers use social networking sites with their students?
- Do you use social networking sites to interact with your lecturers?

The responses of respondents whose teachers had given them access to their social media accounts are shown in Figure 7.1. There are three categories for the responses: yes, to some extent, and no. 38.6% of those surveyed deny that any academics ever granted them access to their social media accounts. But according to 31% of students, some of their professors have granted them access to their social media pages.

Only 30.4% of the participants in the survey say they have access to every professor's social networking account. In order to determine whether faculty members at their university are activating and using their social media accounts to engage with their students, the figures 8.1 and 9.1 present these data. The findings show that 37.3% of respondents claimed that professors do not engage with them on social media. The fact that 34.8 percent of respondents said they do not contact with their teachers via social media supports this assertion.

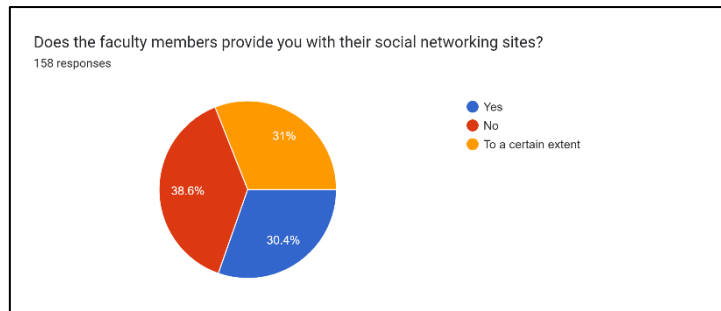


Figure 7.1 Educators allowing pupils access to their social media profiles

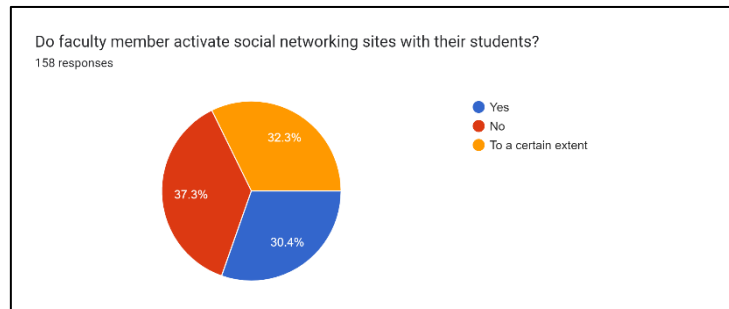


Figure 8.1 Faculty's active student social networking sites

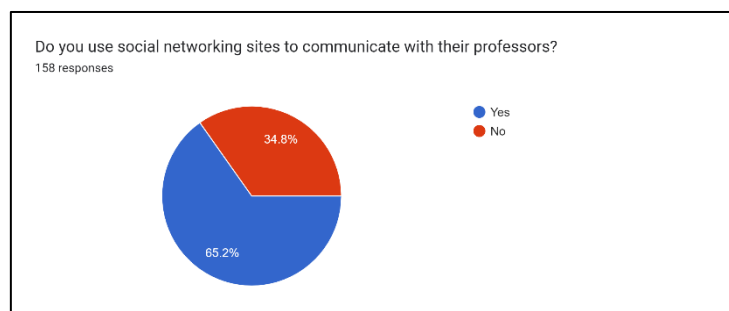


Figure 9.1 Social networks enable communication between teachers and students.

3.4. R4 – How the social media networks can improve the educational process during the lecture?

The effectiveness of social media utilization during the lecture as an educational operation was evaluated using three questions.

The questions are as follows:

- Does using faculty members as a kind of social communication during the educational process help students' academic success?

- Do you think it has an impact on a student's comprehension of the presentation if they use social media during a lecture while pretending to be doing academic work?
- Do you favor allowing students to use social media during class? The findings reveal that 70% of respondents (32.3 percent agree and 40.5 percent are neutral) are not against faculty members using social media during lectures and that this use positively influences students' academic performance, as shown in figures 10.1 and 10.2.

However, the findings show that respondents do not agree with students utilizing social media during lectures for academic study; just 39.9% of respondents agree with this statement, while the remaining 39.9% are indifferent, and 20.3 % of respondents disagree.

These findings are illustrated in Figures 11.1 and 11.2

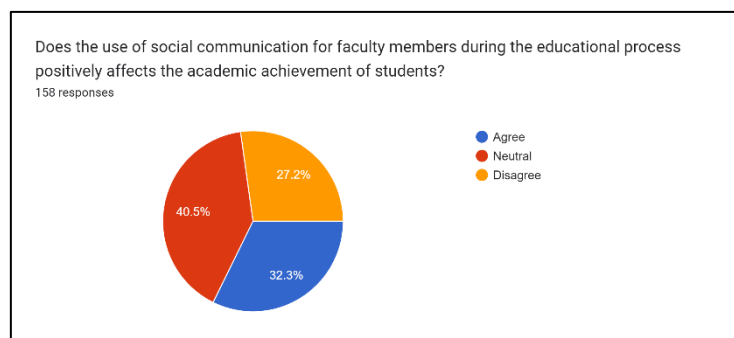


Figure 10.1 the use of social media by lecturers during class

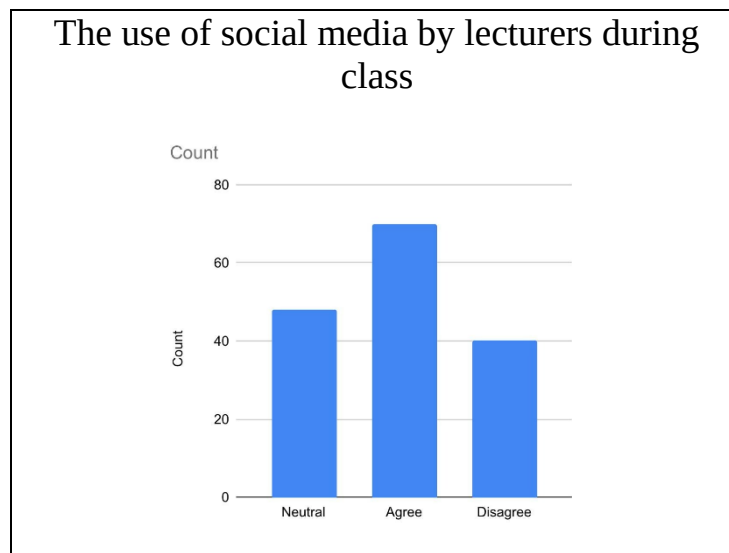


Figure 10.2 the use of social media by lecturers during class

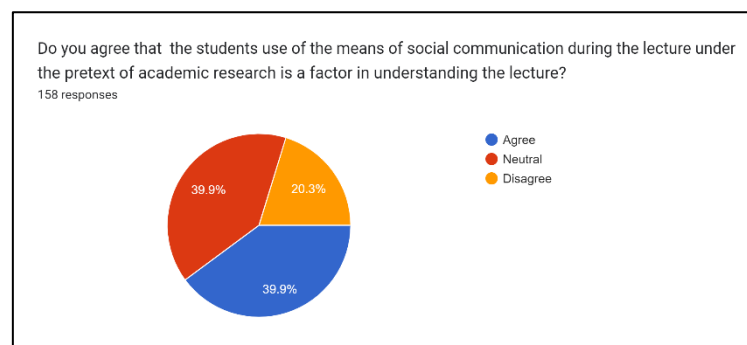


Figure 11.1 students utilizing social media during lectures for academic purposes

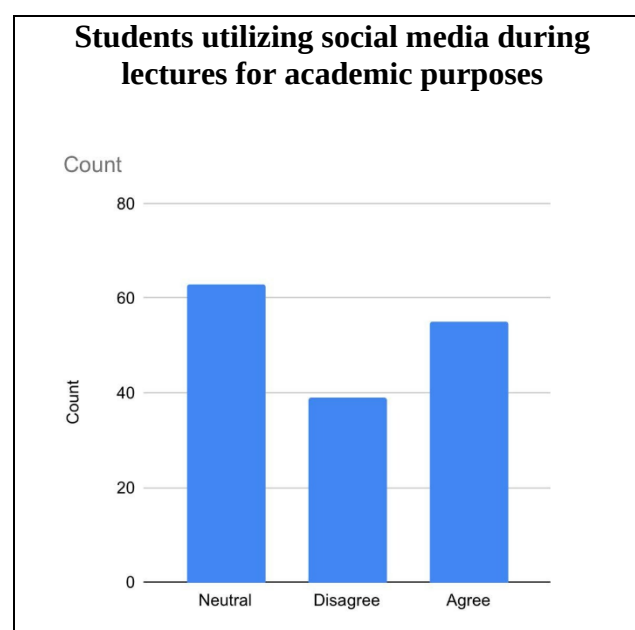


Figure 11.2 students utilizing social media during lectures for academic purposes

We generally questioned the respondents if they agreed with the usage of social media by students or teachers during class lectures. Figures 12.1 and 12.2 of the data show that 35.4 percent of respondents oppose using social media during lectures, 31.6 percent of respondents agree with this statement, and 35.4 percent of respondents are undecided.

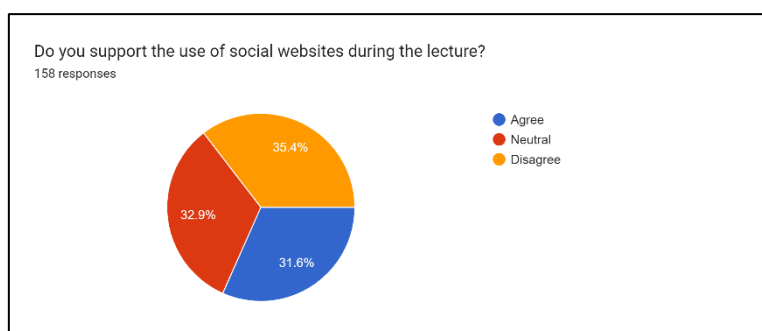


Figure 12.1 Social media usage during lectures

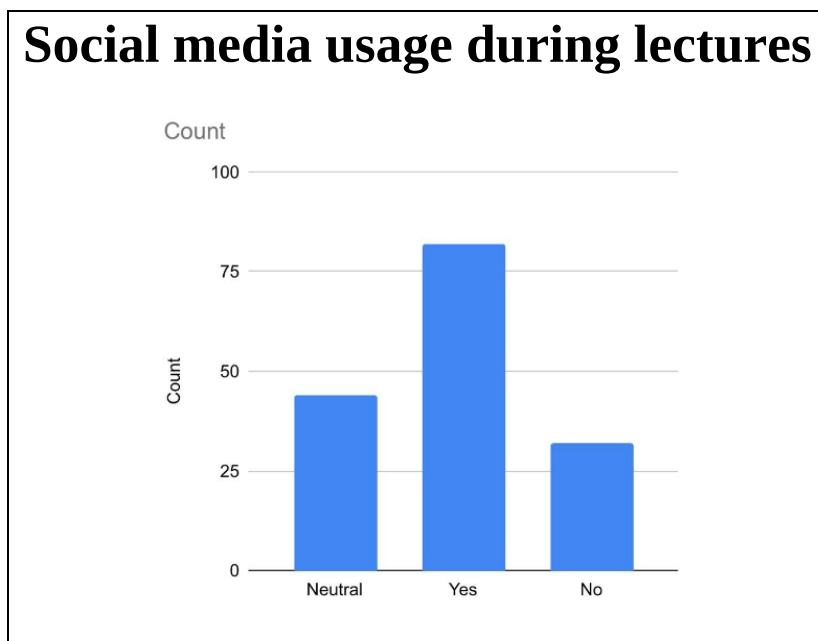


Figure 12.2 Social media usage during lectures

3.5 R5 – Does using social media networks in the classroom have any advantages?

The effectiveness of social media networks in advancing the educational process has been investigated using four questions.

The questions are as follows:

- Can the instructor accurately impart the required knowledge via social networking sites?
- Do you think that social networking sites have a substantial impact on how well pupils succeed in school?
- Do you think social networks have enhanced faculty communication but not academic quality of studies?
- Which social media platforms are most useful for the learning process?

The statistics show that 44.3% of respondents concur that using social networks, a teacher can quickly confirm the accuracy of information. Additionally, 30.4% of respondents are neutral, but 25.3% of them concur that the teacher did not adequately communicate the required information via social media. These facts are illustrated in Figure 13.1.

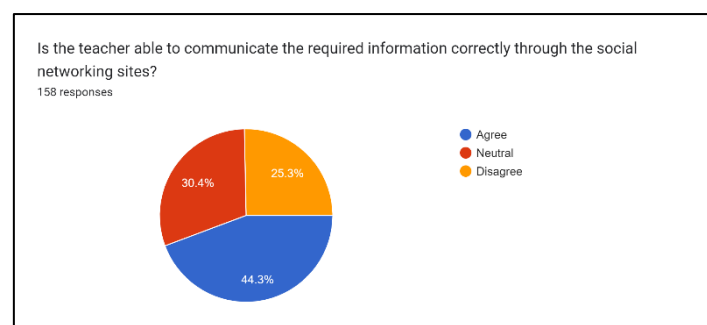


Figure 13.1 the dissemination of reliable information through social media

35% of respondents agreed that using social media helps students perform better academically. A further 40.1 percent, however, expressed their lack of opinion, and 24.8 percent denied that social media improved academic performance in kids. The various replies are depicted in Figure 14.1.

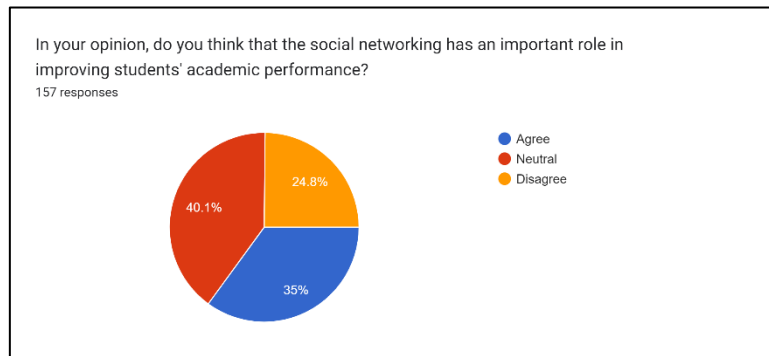


Figure 14.1 Utilizing social networking to boost students' academic performance

Figure 15.1 demonstrates that 51.9 percent of respondents agreed with the assertion that social media had improved communication between students and teachers but had not raised the bar for academic accomplishment, while 27.8 percent were indifferent and 20.3 percent disagreed. At the end of the questionnaire, we asked the students which social network they believed would most benefit the educational process. YouTube is the best website for boosting learning, followed by Twitter, Facebook, and other social media sites like LinkedIn, Instagram, and WhatsApp. YouTube is the best social network for boosting the educational process, according to 46.8% of the 158 respondents, while Facebook is the best social network for enhancing the educational process, according to 27.8% of the respondents. Twitter community. The outcomes are displayed in figure 16.1.

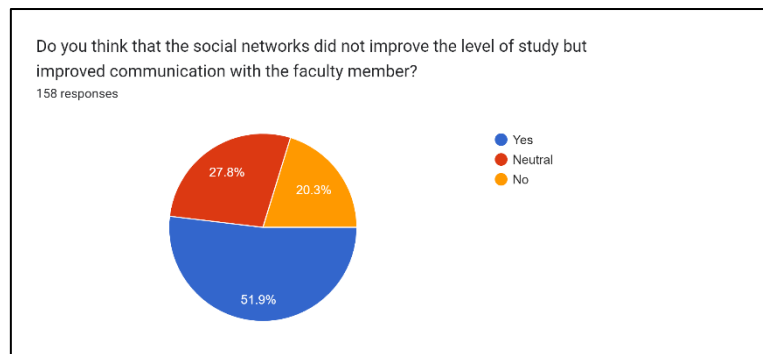


Figure 15.1 Social networks promote communication with faculty members, not the quality of study.

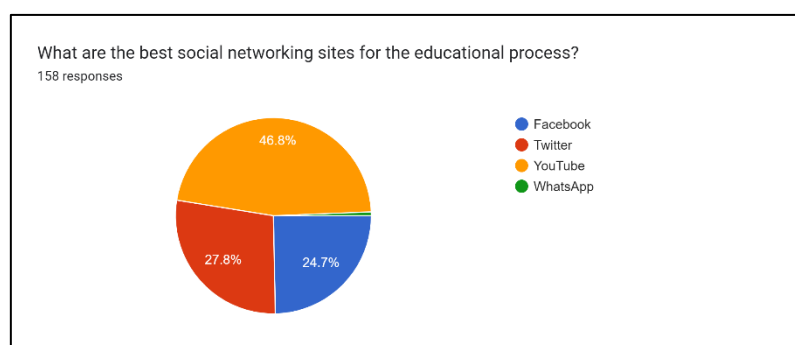


Figure 16.1 The top social networks for education

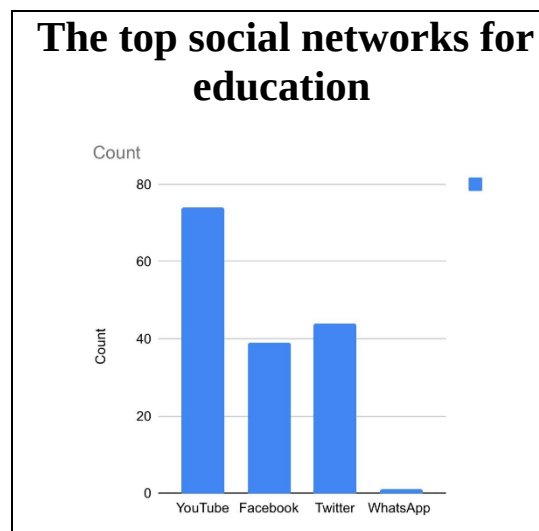


Figure 16.2 The top social networks for education

CHAPTER 4 CONCEPT AND CALCULATION OF CHI-SQUARE

Structure

4.0 Introduction

4.1 Objectives

4.2 Application of Chi-square test

4.2.1 Chi-square test of goodness of fit

4.2.2 Testing hypothesis

4.2.3 Chi-square test as a test of independence

4.0 INTRODUCTION

When we have data in the form of frequencies, ordinal levels of measurement, or nominal levels of measurement, we employ this. There are several uses for the chi-square test. You were given a conceptual understanding of the chi-square test in the previous unit, and in this one, we'll talk about how to compute the chi-square.

4.1 OBJECTIVES

After reading this section you will be able to:

- Identify Chi-Square as a test of goodness of fit;
- Calculate Chi-Square as a test of independence.

4.2 APPLICATION OF CHI- SQUARE

The Chi-square test is used for comparing experimentally obtained results with those to be expected. The important application of Chi-square is as given below:

4.2.1 Chi-square as a Test of Goodness of Fit

To determine whether the observed value of a particular phenomena differs considerably from the expected value, the Chi-Square goodness of fit test is utilized. The phrase "goodness of fit" is used in the Chi-Square goodness of fit test to compare the expected probability distribution to the observed sample distribution. How well a theoretical distribution (such the normal, binomial, or Poisson) fits the empirical distribution is determined by the Chi-Square goodness of fit test. Intervals are created from the sample data for the Chi-Square goodness of fit test.

Following that, the observed and actual numbers of points inside each period are compared.

Establish the hypothesis for the Chi-Square goodness of fit test, which is based on both the null and the alternative hypothesis, with regard to the process.

a) Null hypothesis: The null hypothesis states that there is no discernible difference between the observed and expected result in the Chi-Square goodness of fit test.

b) Alternative hypothesis: In the Chi-Square goodness of fit test, the alternative hypothesis asserts that there is a significant difference between the observed and expected value. Using the calculation, determine whether the chi-square value is significant at the.05 or.01 levels for the specified degrees of freedom. Reject the null hypothesis in that case. Accept the null hypothesis if not significant.

4.2.2 Testing Hypothesis Of Equal Probability

The Chi-square test is a practical tool for contrasting experimental results with those that would be predicted theoretically based on a given hypothesis. The equation to determine χ^2 is

$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Concept and Calculation of Chi Square

Where;

O_i = observed frequency of a phenomenon or even which the experimenter is studying

E_i = expected frequency of the same phenomenon based on "no difference" or "null" hypotheses.

The Chi-square test is a practical tool for contrasting experimental results with those that would be predicted theoretically based on a given hypothesis. The equation to determine χ^2 is

The following example can be used to demonstrate how to apply the aforementioned formula.

Example

The total number of replies from all social media sites are shown in the survey. Of the 158 respondents, 59 utilize YouTube and 41 use Instagram to communicate with one another. Instagram is the next-most popular social network for student communication, after YouTube.

The first row of table 2's first column contains the observed data (O_i).

1. If every answer was chosen equally, the distribution of answers in the second row is what one would anticipate based on the null hypothesis (E_i).

Table 4.1 Responses from subjects regarding the social media used

S. NO.	Social media used	Observed (O)	Expected (E)	(O-E)	(O-E) ²	(O-E) ² /E
	Insta	41	39.5	1.5	2.25	0.05
	Facebook	25	39.5	-14.5	210.25	5.32
	Twitter	33	39.5	-6.5	42.25	1.06
	YouTube	59	39.5	-10.5	110.25	2.79
	Total	158				

No. of levels (k) = 4

$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$X^2 = \sum 365/158$$

$$X^2 = 2.31$$

X^2 = Calculated chi square= 2.31

Degree of freedom= No. of levels -1 (k)

Degree of freedom= 4.1

Degree of freedom = 3

Entering table of X^2 , we find in row d.f = 3 a value 7.21 given under the heading

Levels of significance respectively

Chi-square Distribution Table

d.f.	.995	.99	.975	.95	.9	.1	.05	.025	.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57
22	8.64	9.54	10.98	12.34	14.04	30.81	33.92	36.78	40.29
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98
26	11.16	12.20	13.84	15.38	17.29	35.56	38.89	41.92	45.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89
32	15.13	16.36	18.29	20.07	22.27	42.58	46.19	49.48	53.49
34	16.50	17.79	19.81	21.66	23.95	44.90	48.60	51.97	56.06
38	19.29	20.69	22.88	24.88	27.34	49.51	53.38	56.90	61.16
42	22.14	23.65	26.00	28.14	30.77	54.09	58.12	61.78	66.21
46	25.04	26.66	29.16	31.44	34.22	58.64	62.83	66.62	71.20
50	27.99	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15
55	31.73	33.57	36.40	38.96	42.06	68.80	73.31	77.38	82.29
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38
65	39.38	41.44	44.60	47.45	50.88	79.97	84.82	89.18	94.42
70	43.28	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.43
75	47.21	49.48	52.94	56.05	59.79	91.06	96.22	100.84	106.39
80	51.17	53.54	57.15	60.39	64.28	96.58	101.88	106.63	112.33
85	55.17	57.63	61.39	64.75	68.78	102.08	107.52	112.39	118.24
90	59.20	61.75	65.65	69.13	73.29	107.57	113.15	118.14	124.12
95	63.25	65.90	69.92	73.52	77.82	113.04	118.75	123.86	129.97
100	67.33	70.06	74.22	77.93	82.36	118.50	124.34	129.56	135.81

Our obtained value in 2.31, which is far above the given value in the table

If calculated chi-square value $\leq$ the value in table 2.2

We are unable to disprove the null hypothesis. But from the results it may be stated quite confidently that the value of calculate chi-square is less than critical value.

Consequently, we are unable to reject the null hypothesis.

4.2.3 Chi square test of independence

A specific version of Pearson's chi-square test is the chi-square (χ^2) test of independence. Chi-square tests by Pearson are nonparametric assessments of categorical variables. They are employed to ascertain whether your data differ considerably from your expectations. To ascertain if two categorical variables are connected, you can perform a chi-square test of independence, also referred to as a chi-square test of association. When two variables are connected, the likelihood that one will have a particular value depends on the value of the other variable.

The observed frequencies, or the number of observations in each combined group, are used in the chi-square test of independence calculations. In order to determine if two variables are unrelated, the test compares observed frequencies to those you would anticipate. Similarities will exist between observed and anticipated frequencies.

The chi-square of independence assesses the null and alternative hypotheses, just like all hypothesis tests.

H_0 : Men and women both prefer to use social media during specific times or hours each day.

H_a : Neither men nor women favor using social media during particular hours of the day or for a specific period of time each day.

There are 91 men and 67 women listed in the third double entry, also known as a two-way double entry, in the following table.

Table 4.2: Frequencies of respondents in terms of view expressed

Gender	Agree	Neutral	Disagree	Total
Male	29	30	31	91
Female	14	32	22	67
Total	43	62	53	158

As mentioned above, we give the data in contingency table format for this. The computed anticipated frequencies are listed in brackets next to the corresponding observed frequencies. Table 4.3 provides the contingency table and the method for calculating predicted frequencies.

Table 4.3: Contingency table with O_i and E_i for the given data given in table 4.2

Gender	Agree	Neutral	Disagree	Total
Male	29 (30.3)	30 (30.3)	31 (30.3)	91
Female	14 (20.3)	32 (20.3)	22 (20.3)	67
Total	43	62	53	158

The method to determine each cell's anticipated frequency in a particular cell:

$$43 \times 91/158 = 67.4 \qquad 62 \times 91/158 = 35.7 \qquad 53 \times 91/158 = 30.52$$

$$43 \times 67/158 = 18.23 \qquad 62 \times 67/158 = 26.29 \qquad 53 \times 67/158 = 22.47$$

The standard formula can be used to calculate the value of X^2 ,

$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

O_i	E_i	$(O_i - E_i)$	$(O_i - E_i)^2$	$(O_i - E_i)^2$
29	30.3	-1.3	1.69	0.05
30	30.3	-0.3	0.09	0.02
31	30.3	0.7	0.0049	0.00016
14	30.3	6.3	39.69	1.95
32	30.3	11.7	136.89	6.75
22	30.3	1.7	2.09	0.14
				X^2

Table 4.4 Computation of X^2

With the aid of the normal formula, the value of X^2 may be calculated.

For 2 df circuit value at .05 level is 5.99. our obtained value of X^2 is 19.86. It is far higher than the table value. Therefore, we reject the null hypothesis

Steps

Create the null hypothesis first.

2) Use the technique in the table to determine the expected values.

3) Calculate the discrepancy between the observed and expected values for each cell.

4) Square each discrepancy and divide it by the anticipated frequency in each cell,

5) Add these values together, and you get χ^2 as the result.

CONCLUSION

The main objective of this study was to investigate how students use social media and how that use connects to their academic performance. One of the study's flaws is the lack of student interaction. In actuality, the research revealed that the usage of online social media had improved communication between instructors, staff, and students, assisting in the dissemination of factual information and advancing students' understanding of concepts and course material. The facts collected make it clear that most respondents do not recommend utilizing social media during class time. Students claim that Twitter and YouTube are the social media sites that contribute most to education. Given that students utilize social media extensively, more study should be conducted with a focus on how social media influences students' current life rhythms and their social interactions.

RECOMMENDATIONS

It is advised that we refrain from using social media during lectures and that more research be done on the effects of social media use on the connections that students form with one another.

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